

Methods: Candidate_Black_Hole_X-ray_Binary

ALMA Thesis Planner (automated pipeline)

0.1 0. Software environment, bookkeeping, and reproducibility

Everything runs in CASA plus standard Python; nothing else is needed. Before touching data, set up the scaffolding you will use for the whole year, because this thesis is as much a calibration-systematics result as a timing result and the audit trail *is* a deliverable.

Create a git repository with three top-level areas: ‘restore/’ (CASA restoration scripts and logs, one directory per execution block), ‘pipeline/’ (the extraction and analysis code), and ‘products/’ (per-epoch tables and figures). Every epoch is driven by a small YAML configuration file recording the MS name, the CASA version used to restore it, the target field ID, the spectral-window (SPW) grouping, the native integration time, the time bins to be used, and the random seeds for the simulations. No number enters the thesis that cannot be regenerated by running one script against one config file. Use ‘conda/’/‘pip’ environments pinned in a lockfile, and record the CASA build string (‘casalog.version()’) in every log. Analysis code is Python 3 with ‘numpy’, ‘scipy’, ‘astropy’ (including ‘astropy.timeseries.LombScargle’), ‘matplotlib’, and CASA’s ‘casatools’ (‘table’, ‘ms’, ‘msmetadata’) for direct visibility access — you will read visibilities as arrays far more often than you will run CASA tasks.

0.2 1. Step 0 (Weeks 1–3): retrieval, calibration restoration, and the EB audit

Retrieval. Query the ALMA Science Archive for project code 2021.1.01208.T using ‘astroquery.alma’ (‘Alma.query(project_code=...)’), then request the *unprocessed* delivery: the raw ASDMs plus the calibration tables, ‘scriptForPI.py’, and the QA2 weblog. Do not accept only the delivered science-ready products; you need visibilities at native time resolution, and the delivered continuum images are useless for this project. Download per execution block (EB) so that a failure in one does not force a re-fetch of all.

Restoration. Open the weblog first and read off the pipeline/CASA version used for the original reduction. Install exactly that CASA version — restoration is version-brittle, and for a ToO with possible manual calibration the ‘scriptForPI.py’ may call tasks whose defaults have changed. Run ‘scriptForPI.py’ in the ‘calibration/’ directory of each EB to regenerate ‘*.ms.split.cal’. Restoration is confirmed successful, EB by EB, when (i) the script exits without error, (ii) ‘listobs’ on the restored MS shows the expected fields and SPWs, and (iii) a quick ‘tclean’ of the phase calibrator and of the target reproduces the flux densities quoted in the weblog to within a few percent. Log every deviation. Budget the full two weeks; if a script fails, the usual causes are a missing antenna-position table, a Tsys/WVR table path assumption, or a CASA minor-version difference — work these one at a time and record the fix.

The audit. For each restored EB, tabulate — using ‘listobs’, ‘asdmsummary’, ‘msmetadata’, and direct reads of the ‘MAIN’ table — the following: UTC start/stop and elapsed time; the **native**

integration time (read the ‘EXPOSURE’ and ‘INTERVAL’ columns directly; do not trust a nominal value, and understand that the MS cannot be rebinned finer than what was written); the scan pattern (target/phase-calibrator cycle time and the number and length of target scans); total on-source time; the number of antennas and the array configuration / baseline-length range; **whether a check source was observed**; the spectral setup (number of SPWs, bandwidth, channelization, sideband placement and therefore the frequency lever arm); the flagged fraction per antenna, per SPW, and per scan; the elevation, airmass, and PWV track (from the ‘WEATHER’ and ‘POINTING’/‘SOURCE’ subtables and the weblog); and the mean target flux density from a full-EB ‘tclean’ image plus the image rms.

From this table, and using image-plane and uv -plane checks (a per-EB image of the target field to identify any other sources needing subtraction, and a plot of amplitude versus uv -distance to confirm the target is unresolved), you and I sign off in week three on which branch is in play:

- **EB count:** ≥ 5 usable EBs $\rightarrow$ the full Step 5 evolution study; 3–4 $\rightarrow$ Step 5 reduces to a two-point comparison; 1–2 $\rightarrow$ Step 5 is dropped and the thesis becomes a single-epoch deep study on the best block.
- **Check source present or absent** $\rightarrow$ whether external floor estimator (c) is available.
- **Target brightness:** tens of mJy $\rightarrow$ detections and measured fractional rms and PSD slope; few mJy $\rightarrow$ the same machinery delivering a floor plus upper limits.

The signed-off branch, including the explicit possibility that the guaranteed result is a systematics floor plus upper limits, is written into the thesis plan in month one. The audit table is deliverable (i).

0.3 2. Step 1: Visibility-domain light-curve extraction

Work in the visibility domain throughout. Per-time-bin imaging wastes photons on deconvolution and injects CLEAN-dependent artefacts at exactly the amplitude level you are trying to characterize.

For each EB, ‘split’ the target field from the restored MS with ‘datacolumn=’corrected’, ‘keep-flags=False’, and time averaging left at native resolution. Average in frequency only within each SPW (or within each sideband, retaining the two sidebands separately so Step 4’s sideband check is possible), and only after confirming from the weblog that no channel-dependent flagging would bias the average. Recompute visibility weights with ‘statwt’ using a ‘timebin’ chosen to be *longer* than the longest variability timescale you will study but short enough to track atmospheric changes — otherwise source variability leaks into the weights. Cross-check the resulting weights against the empirical scatter of the real part of the visibilities across baselines within a bin; the two should agree to $\sim 10\%$, and if they do not, use the empirical values.

Phase-rotate to the target position with ‘fixvis’, subtract any confusing field sources identified in Step 0 (‘tclean’ a model excluding the target, then ‘ft’ + ‘uvsub’), and then extract the light curve: in each time bin, fit a point source at the phase centre by forming the weighted mean of the real part of the visibilities (equivalently ‘uvmodelfit’ with ‘comptype=’P’ fixed at the origin), with the uncertainty from the weights and cross-checked against the weighted scatter of the imaginary part, which is a pure noise estimate. Do this at the native integration time and at logarithmically spaced rebinnings — roughly factors of two — up to the target-scan length. Store each light curve as an ‘astropy’ table with columns: time, flux, thermal σ , number of visibilities, number of antennas

present, flagged fraction, elevation, airmass, PWV.

Self-calibration on the target, if the SNR permits it at all, is **phase-only** and on solution intervals *longer* than any timescale under study, with ‘combine=’spw’. Amplitude self-calibration is forbidden: it would absorb the signal into the gain table. Enforce this in code — the extraction script asserts ‘calmode == ’p’ and that ‘solint’ exceeds the longest analysed timescale, and refuses to run otherwise.

0.4 3. Step 2: The systematics floor (the core of the thesis)

The phase calibrator’s corrected light curve is flat almost by construction — the gains were derived from it — so it must never be used as the floor. Build the floor from three independent estimators, and be precise about which component of the error budget each one bounds.

(a) Closure amplitudes on the target. For every antenna quadrangle (i,j,k,l) in each time bin, form $A_{ijkl} = (|V_{ij}||V_{kl}|)/(|V_{ik}||V_{jl}|)$ and work with $\log A$, which is better behaved. Select a maximally independent set (there are $N(N-3)/2$ independent closure amplitudes for N antennas) using a fixed reference-antenna construction so the set is reproducible across bins; propagate thermal errors analytically and apply the standard low-SNR debiasing, requiring per-baseline SNR above roughly 5 in each bin — this will often force a coarser bin for the closure analysis than for the flux light curve, which is fine and must be stated. Closure amplitudes are immune to *antenna-based* gain errors by construction, including per-antenna atmospheric amplitude decorrelation. They therefore give you two things directly and cleanly: a test that the single-point-source model underlying Step 1 holds bin by bin (for an unresolved source, $\log A$ is consistent with zero within thermal noise), and a measurement of any *baseline-based* systematics or genuine structural evolution, as excess scatter in $\log A$ above the thermal expectation. Quote that excess, as a function of timescale, as the structural/baseline-based term $\sigma_{\text{struct}}(\tau)$. Because it is blind to antenna-based errors, it does not by itself bound the antenna-based flux-scale term — that is what (b) and (c) are for, and the three are combined below. Compute the closure phases as well and check them for consistency with zero; a closure-phase excess is the most sensitive alarm that the point-source assumption is failing.

(b) Gain cross-validation, always. Split the phase-calibrator scans into alternating subsets. Run ‘gaincal’ on subset A only, ‘appylcal’ those solutions (interpolated) to the held-out subset B scans, and measure the fitted calibrator flux in B. The scatter of the held-out fluxes about their mean is a direct, empirical measurement of the gain-transfer error. Repeat with A and B exchanged, and repeat while varying the effective scan separation (skip one scan, two scans, ...) to obtain $\sigma_{\text{gain}}(\Delta t)$ as a function of the calibrator–target interpolation gap. Also vary ‘solint’ to map how the transfer error depends on solution interval. This is the estimator that bounds the antenna-based term at the timescales that matter, and it is always available.

(c) Check source, if Step 0 found one. This is the clean external case: run it through the identical Step 1 extraction and Step 3 statistics, treating it exactly as the target. Its measured fractional rms *is* the floor at each timescale, with no modelling assumptions.

Cross-checks, not evidence. Antenna-drop jackknives (recompute the light curve with each antenna removed in turn) and split-half baseline jackknives (short versus long baselines) are run and reported, but only as consistency checks: in a compact configuration neighbouring antennas see much of the same troposphere, so the “non-common-mode” premise partly fails and these tests

are optimistic.

Long timescales. On hour-long timescales the interleaved calibrator sampling gives little leverage, so supplement with regression: fit the per-bin residual flux (light curve minus a smooth trend) and the per-antenna gain amplitude statistics against elevation, airmass, PWV, and flagged fraction, using ordinary least squares with bootstrap errors. Any significant coefficient is converted into an equivalent fractional-rms contribution over the epoch and added to the floor. Explicitly check whether antennas entering or leaving the solution (through flagging) coincide with apparent flux steps.

Output. Combine in quadrature into a single fractional-rms systematic floor as a function of timescale, $\sigma_{\text{sys}}(\tau) = [\sigma_{\text{gain}}^2(\tau) + \sigma_{\text{struct}}^2(\tau) + \sigma_{\text{env}}^2(\tau)]^{1/2}$, with $\sigma_{\text{check}}(\tau)$ reported alongside as the external validation where available. Plot it. Target variability is claimed astrophysical only where it exceeds this curve, and the curve is carried through every subsequent step, including the simulations.

0.5 4. Step 3: Ensemble timing statistics

Fractional rms. For each epoch and each timescale, compute the excess variance, $\sigma_{\text{XS}}^2 = S^2 - \langle \sigma_{\text{err}}^2 \rangle$, where S^2 is the sample variance of the light curve and σ_{err} the per-bin thermal error, and $F_{\text{var}} = \sigma_{\text{XS}} / \langle F \rangle$, with uncertainties from the standard Vaughan et al. (2003) prescriptions and cross-checked by bootstrap. Then subtract the Step 2 floor in quadrature and propagate its uncertainty. Plot F_{var} versus timescale (the rms spectrum), computed in segments of fixed length so segments are comparable. Where the floor-corrected value is not significant, quote a one-sided upper limit at 95% confidence, computed from the simulations below rather than analytically.

Power spectral density. Because phase-referenced observing imposes hard, near-periodic gaps whose window function aliases power, do not fit a naive periodogram. Compute a Lomb–Scargle periodogram on the actual (gappy) time stamps, in fractional-rms-squared normalization, over a frequency range quoted explicitly from the Step 0 audit — roughly $1/(2 \times \text{native integration time})$ down to $1/(\text{scan length})$, about 1.5 decades. Fit a **slope only**, $P(f) = A f^{-\beta} + C$, with C the thermal white-noise level fixed to $2 \langle \sigma_{\text{err}}^2 \rangle \Delta t$. No break fitting: the dynamic range does not support it.

Fit by forward modelling, using the PSRESP-style procedure. Generate large ensembles (≥ 2000 per trial slope) of red-noise light curves with the Timmer & König (1995) algorithm on a much finer grid and a much longer duration than the data (to capture red-noise leak and aliasing), resample them onto the **exact observed time stamps**, apply the **actual flagging pattern**, add per-bin Gaussian thermal noise at the **measured** σ_{err} , and multiply by simulated gain perturbations drawn to reproduce the Step 2 floor’s amplitude and its temporal correlation structure. Periodogram each realization identically to the data. Scan β on a grid, form the χ^2 -like distance between the observed periodogram and the ensemble mean in log space using the ensemble spread as the error, and take the best-fit β and its confidence interval from the distribution of that statistic across simulations. Report the goodness-of-fit p-value; a poor fit at all β is itself a result. Before touching the data, validate the whole chain by injecting known slopes and confirming unbiased recovery — this is the slope-recovery test, and its figure goes in the thesis.

rms–flux relation. Divide each light curve into segments; for each segment compute the floor-corrected excess rms and the mean flux; fit a straight line and compute the Spearman coefficient. A linear rms–flux relation indicates multiplicative, propagating-fluctuation variability inherited from the accretion flow; its absence favours additive, discrete-ejection variability. Calibrate the null with the same simulations: a stationary Gaussian red-noise process *without* rms–flux coupling can produce a spurious positive correlation at these segment counts, so quote significance against the simulated distribution, not against a nominal p-value.

0.6 5. Step 4: The event test as a falsification of the Step 3 model

If the variability is red noise, "flare rate" and "fraction of flux in discrete events" are threshold-dependent restatements of the Step 3 slope and normalization. So pose the census as a test of the Step 3 model, using the simulation ensemble already built at the best-fit slope and normalization, passed through the real sampling, flagging, noise, and Step 2 floor.

Ask exactly two questions. **(i) Excess of high-significance excursions.** Define excursions with a fixed, pre-registered detection statistic — a matched filter against a family of exponential-rise/exponential-decay kernels spanning the accessible timescales, with the detection threshold expressed in units of the local combined thermal-plus-floor uncertainty. Run it identically on data and on every simulated realization. Compare the observed number of excursions above each threshold with the simulated distribution; report the p-value as a function of threshold, and correct for the number of thresholds tried by pre-registering the grid. **(ii) Rise/decay asymmetry.** For the detected excursions, measure the asymmetry with a shape statistic computed the same way on data and simulations — for example the ratio of fitted rise to decay e-folding times, or the skewness of the light-curve first differences within the excursion window. Compare to the null distribution from the stationary realizations.

Only excursions that fail the simulated null are reported as discrete-ejection candidates. For those, fit the FRED profile and report amplitude, rise time, decay time, and fluence, with uncertainties from refitting bootstrap resamples of the visibilities in the event window; these are the numbers that constrain the size and replenishment rate of the emitting region. If nothing exceeds the null, state plainly that the light curves are consistent with a single stationary stochastic process of the measured slope, and report the corresponding upper limit on the excursion rate from the simulations.

Sideband check. Fit the flux independently in the lower- and upper-sideband SPW groups, form the ratio per bin, and verify it is consistent with unity at the precision the Step 2 floor independently establishes. Report this in one sentence. Do not attempt spectral-index-versus-time science; the fractional frequency lever arm is far too small to support it.

0.7 6. Step 5: Evolution across the outburst

Repeat Steps 1–4 for every usable epoch, with the pipeline driven from each epoch's config file so extraction, binning, thresholds, and simulation settings are identical by construction. Because the antenna count, configuration, and weather differ between EBs, **recompute the floor per epoch** — never carry one epoch's floor into another.

Order the epochs by mean sub-mm flux density, which serves as the proxy for outburst stage, and plot against it: (a) the systematic floor at two or three fiducial timescales; (b) floor-corrected fractional rms or its upper limit; (c) PSD slope with its forward-modelled confidence interval; (d)

rms–flux relation slope; and (e) any null-failing event properties. Test for monotonic trends with a Spearman rank correlation across epochs and quote the coefficient and its bootstrap interval, given the small number of points. This delivers the “as a function of accretion rate” ambition in the one band you control. Under the 3–4 EB branch this becomes a two-point comparison, reported as a difference with a propagated uncertainty rather than a trend; under the 1–2 EB branch this step is not attempted and effort goes into the single-epoch deep study fixed in month one.

0.8 7. Schedule

Weeks 1–3: retrieval, restoration, audit table, branch sign-off. Weeks 4–8: Step 1 extraction pipeline, validated by recovering the weblog flux density from the full-EB light-curve mean. Weeks 9–16: Step 2 floor, all three estimators plus regressions and jackknife cross-checks; this is where most of the year’s intellectual weight sits. Weeks 17–24: Step 3, including building and validating the red-noise-through-real-sampling simulator before any slope is fitted to data. Weeks 25–30: Step 4. Weeks 31–36: Step 5, figures, and writing.

0.9 8. Deliverables

(i) The restoration log and the EB/scan-structure audit table; (ii) the reusable visibility-domain extraction pipeline, with the closure-amplitude and gain-cross-validation floor estimators and the red-noise-through-real-sampling simulator, all seed-controlled and config-driven; (iii) a per-epoch table of systematic floor, floor-corrected fractional rms or upper limit, PSD slope, rms–flux relation, and event-test outcome, with honest uncertainties; (iv) the figures of these quantities versus outburst stage. All ALMA and NRAO acknowledgements as specified in the project documentation are carried in every product.